INTRODUCTION TO C++

WORKSHEET 1

Submitted By: **Sadhabi Dev**
Level : 4
Student ID : 23085140
Cyber Security and Digital Forensics

QUESTION NO.1

Write a program that takes a temperature value from the user. It should then allow the user to choose between Celsius (C) and Fahrenheit (F) for conversion. After the user selection, it should then convert the entered temperature to the chosen scale and display the result.
Use appropriate data types for temperature and handle error like non-numeric input.
Use the following formula for conversion:
F = (C x 9/5) + 32
C = (F - 32) x 5/9

SOLUTION:

```
#include <iostream>
using namespace std;

int main() {
   float c, f, temp;
   int choice;

   cout << "----------------------------" << endl;
   cout << "| Choose the conversion type:|" << endl;
   cout << "----------------------------" << endl;
   cout << "| 1. Celsius to Fahrenheit   |" << endl;
   cout << "| 2. Fahrenheit to Celsius   |" << endl;
   cout << "----------------------------" << endl;
   cout << "Enter your choice (1 or 2):" << endl;
   cin >> choice;

   if (choice == 1) {
      cout << "Enter the temperature in Celsius:" << endl;
      cin >> temp;
      f = (temp * 9.0 / 5.0) + 32;
      cout << temp << " Celsius is " << f << " Fahrenheit." << endl;
   }
   else if (choice == 2) {
      cout << "Enter the temperature in Fahrenheit:" << endl;
      cin >> temp;
      c = (temp - 32) * 5.0 / 9.0;
      cout << temp << " Fahrenheit is " << c << " Celsius." << endl;
   }
   else {
      cout << "Invalid choice!" << endl;
   }

   return 0;
}
```

a. Output of Celsius to Fahrenheit conversion.

```
C:\Users\devsa\OneDrive\Doc
-----------------------------
| Choose the conversion type:|
-----------------------------
| 1. Celsius to Fahrenheit   |
| 2. Fahrenheit to Celsius   |
-----------------------------
Enter your choice (1 or 2):
1
Enter the temperature in Celsius:
89
89 Celsius is 192.2 Fahrenheit.

Process returned 0 (0x0)   execution time : 10.077 s
Press any key to continue.
```

b. Output of Fahrenheit to Celsius conversion.

```
C:\Users\devsa\OneDrive\Doc
-----------------------------
| Choose the conversion type:|
-----------------------------
| 1. Celsius to Fahrenheit   |
| 2. Fahrenheit to Celsius   |
-----------------------------
Enter your choice (1 or 2):
2
Enter the temperature in Fahrenheit:
120
120 Fahrenheit is 48.8889 Celsius.

Process returned 0 (0x0)   execution time : 11.448 s
Press any key to continue.
```

QUESTION NO.2

Write a C++ program to implement a number guessing game with different difficulty levels. Easy difficulty ranges from 1-8, medium from 1-30, hard from 1-50.Then,generate a random number to check if the guess is correct based on the user's selection.

SOLUTION:

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main() {
  int lvl, num, guess, attempt = 5;
  int correct = 0;


  srand(time(0));

  cout << "=====================================" << endl;
  cout << "  * Welcome to the Guessing Game *  " << endl;
  cout << "                                  " << endl;
  cout << "            Enjoy!!!              " << endl;
  cout << "=====================================" << endl;

  cout << " Choose your Difficulty Level " << endl;


  cout << " 1. Easy " << endl;
  cout << " 2. Medium " << endl;
  cout << " 3. Hard " << endl;
  cout << " Enter your choice: " << endl;
  cin >> lvl;

  if (lvl == 1) {
    num = rand() % 8 + 1;
    cout << "\nYou chose Easy!" << endl;
    cout << "\nThe number is between 1 and 8." << endl;
  } else if (lvl == 2) {
    num = rand() % 30 + 1;
    cout << "\nYou chose Medium!" << endl;
    cout << "\nThe number is between 1 and 30." << endl;
  } else if (lvl == 3) {
    num = rand() % 50 + 1;
    cout << "\nYou chose Hard!" << endl;
    cout << "\nThe number is between 1 and 50." << endl;
  } else {
```

```
        cout << "Invalid Choice!" << endl;
        return 0;
    }

    cout << "------------------------------------------" << endl;
    cout << "     Let's start the guessing game!        " << endl;
    cout << "------------------------------------------" << endl;
    cout << "    Try to guess the number!    " << endl;

    while (correct == 0 && attempt > 0) {
        cout << "\nYou have " << attempt << " attempts left." << endl;
        cout << "Enter your guess: ";
        cin >> guess;

        if (guess < num) {
            cout << "Your guess is too low! Try again." << endl;
        } else if (guess > num) {
            cout << "Your guess is too high! Try again." << endl;
        } else {
            cout << "*****************************************" << endl;
            cout << "**   Congratulations! You guessed it!  **" << endl;
            cout << "**                                     **" << endl;
            cout << "**   The correct number is " << num << "!          **" << endl;
            cout << "*****************************************" << endl;
            correct = 1;
        }
        attempt--;
    }

    if (correct == 0) {
        cout << "*******************************************" << endl;
        cout << "**          Sorry! Attempts are Over     **" << endl;
        cout << "**                                       **" << endl;
        cout << "**      The correct number was " << num << "        **." << endl;
        cout << "*******************************************" << endl;
    }

    cout << "================================================" <<
endl;
    cout << "     Thank you for playing the game!          " << endl;
    cout << "================================================" <<
endl;

    return 0;
}
```

The output of number guessing game with difficulties level is given below:

```
C:\Users\devsa\OneDrive\Doc

====================================
  * Welcome to the Guessing Game *

              Enjoy!!!
====================================
 Choose your Difficulty Level
 1. Easy
 2. Medium
 3. Hard
 Enter your choice:
2

You chose Medium!

The number is between 1 and 30.
-------------------------------------------
     Let's start the guessing game!
-------------------------------------------
    Try to guess the number!

You have 5 attempts left.
Enter your guess: 23
Your guess is too high! Try again.

You have 4 attempts left.
Enter your guess: 20
Your guess is too high! Try again.

You have 3 attempts left.
Enter your guess: 15
Your guess is too high! Try again.

You have 2 attempts left.
Enter your guess: 10
Your guess is too high! Try again.

You have 1 attempts left.
Enter your guess: 5
*****************************************
**   Congratulations! You guessed it!  **
**                                     **
**   The correct number is 5!          **
*****************************************
===============================================
      Thank you for playing the game!
===============================================

Process returned 0 (0x0)   execution time : 26.937 s
Press any key to continue.
```

QUESTION NO.3

Write a program that reads an array of integer numbers from the user and sorts the numbers in the ascending order.

SOLUTION:

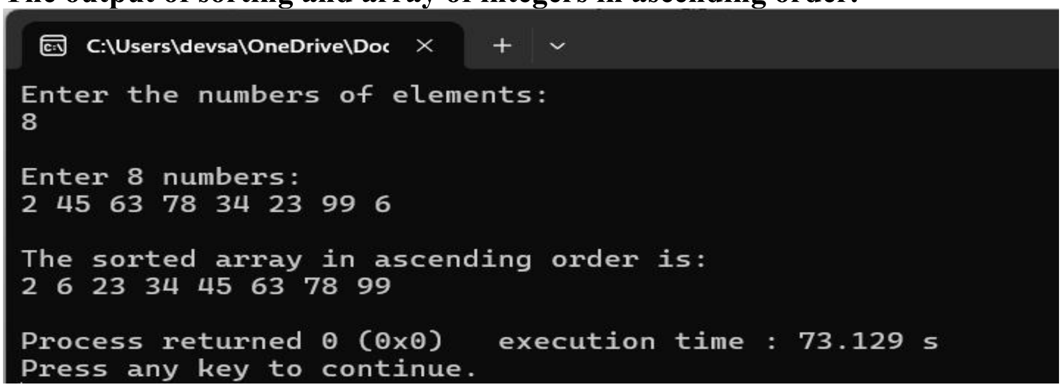
```
#include <iostream>
using namespace std;

int main() {
int a;
cout << "Enter the numbers of elements: "<< endl;
cin >> a;

int arr[a];

cout << "\nEnter " << a << " numbers: " << endl;
for (int i = 0; i < a; i++){
   cin >> arr[i];
}
for (int i = 0; i < a - 1; i++){
for (int j = 0; j < a - 1 - i; j++){
  if (arr[j] > arr[j+1]) {
   int temp = arr[j];
   arr[j] = arr[j + 1];
   arr[j + 1] = temp;
   }
  }
}
cout << "\nThe sorted array in ascending order is:" << endl;
for (int i = 0; i < a; i ++ ){
   cout << arr[i] << " ";
}
cout << endl;
return 0;
}
```

The output of sorting and array of integers in ascending order.



QUESTION NO.4

Write a program that reads a number from the user and based on the user input, it says what day of the week it is, Sundays being 1 and Saturdays being 7. You system should give appropriate response for invalid input entries.

SOLUTION:

```
#include <iostream>
using namespace std;

int main() {
int day;

while (true){
cout << "  \nEnter the number between 1 to 7:  " << endl;
cout << "====================================" << endl;
cin >> day;

if (day == 0)
{
   cout << "====================================" << endl;
   cout << "          Exiting the program"        << endl;
   break;
}

switch(day){
      case 1:
         cout << "* Sunday *" << endl;
         break;
      case 2:
         cout << "* Monday *" << endl;
         break;
      case 3:
         cout << "* Tuesday *" << endl;
         break;
      case 4:
         cout << "* Wednesday *" << endl;
         break;
      case 5:
         cout << "* Thursday *" << endl;
         break;
      case 6:
         cout << "* Friday *" << endl;
         break;
      case 7:
         cout << "* Saturday *" << endl;
         break;
      default:
```

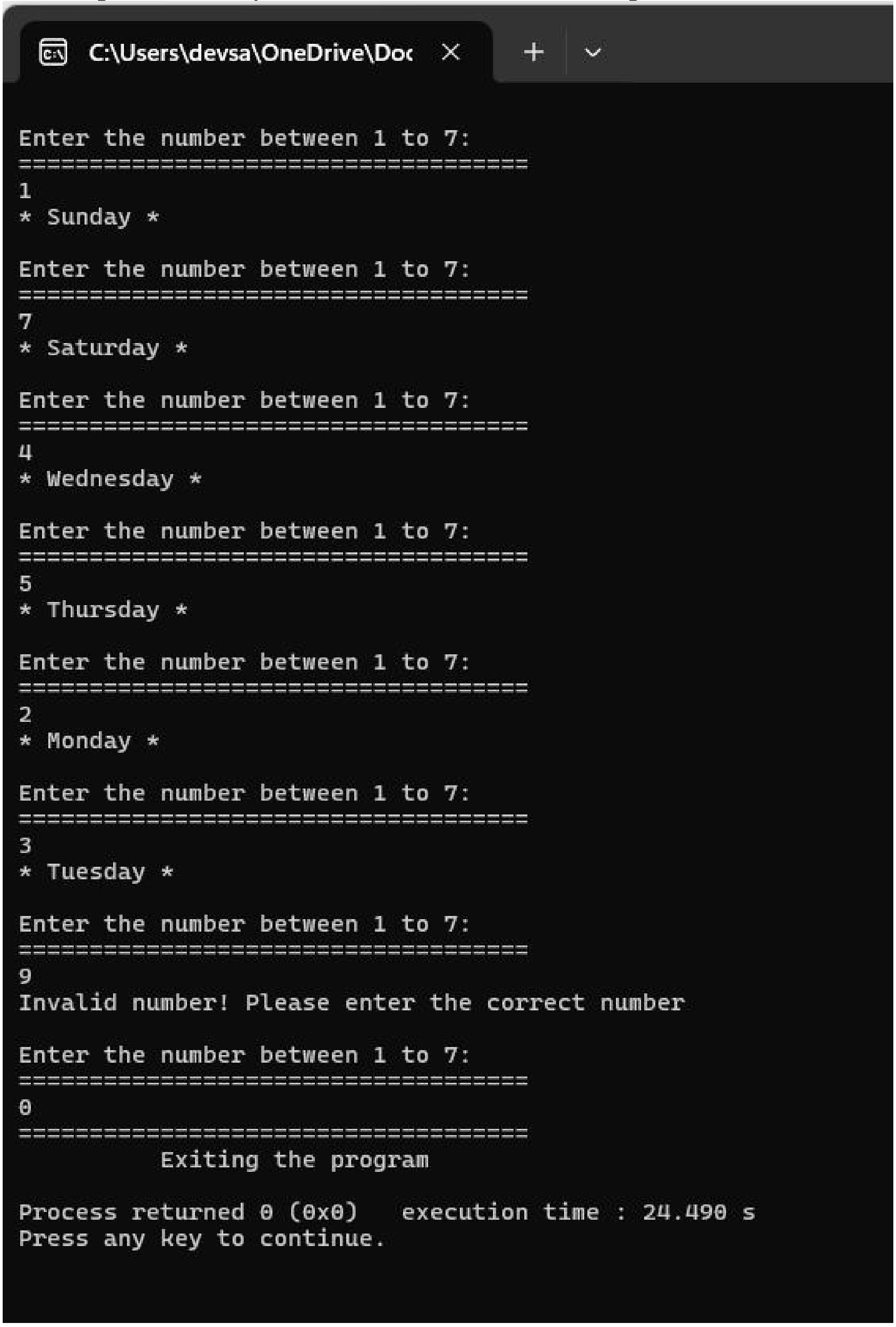
```
        cout << "Invalid number! Please enter the correct number" << endl;
}

}

return 0;

}
```

The output of the days of the week based on user input .



```
C:\Users\devsa\OneDrive\Doc

Enter the number between 1 to 7:
====================================
1
* Sunday *

Enter the number between 1 to 7:
====================================
7
* Saturday *

Enter the number between 1 to 7:
====================================
4
* Wednesday *

Enter the number between 1 to 7:
====================================
5
* Thursday *

Enter the number between 1 to 7:
====================================
2
* Monday *

Enter the number between 1 to 7:
====================================
3
* Tuesday *

Enter the number between 1 to 7:
====================================
9
Invalid number! Please enter the correct number

Enter the number between 1 to 7:
====================================
0
====================================
          Exiting the program

Process returned 0 (0x0)   execution time : 24.490 s
Press any key to continue.
```

QUESTION NO.5

Create a program that takes a positive integer as input and determines whether it's a "**bouncy number**". A bouncy number is one where the digits neither consistently increase nor consistently decrease when read from left to right. For example:

- 123 is NOT bouncy (digits consistently increase)
- 321 is NOT bouncy (digits consistently decrease)
- 120 is bouncy (neither consistently increasing nor decreasing)

SOLUTION:

```
#include <iostream>
using namespace std;

int main() {
    int a;
    char choice;

    do {
        cout << "\nEnter a positive number: " << endl;
        cin >> a;

        int n = a;

        if (a < 10) {
            cout << " " << endl;
            cout << "Single digit numbers are never bouncy." << endl;
        }
        else {
            bool increase = false, decrease = false;
            int lastDigit = a % 10;
            a /= 10;

            while (a > 0) {
                int currentDigit = a % 10;
                if (currentDigit < lastDigit) {
                    decrease = true;
                }
                else if (currentDigit > lastDigit) {
                    increase = true;
                }
                lastDigit = currentDigit;
                a /= 10;
            }

            if (increase && decrease) {
                cout << " " << endl;
                cout << n << " is a bouncy number." << endl;
            }
```

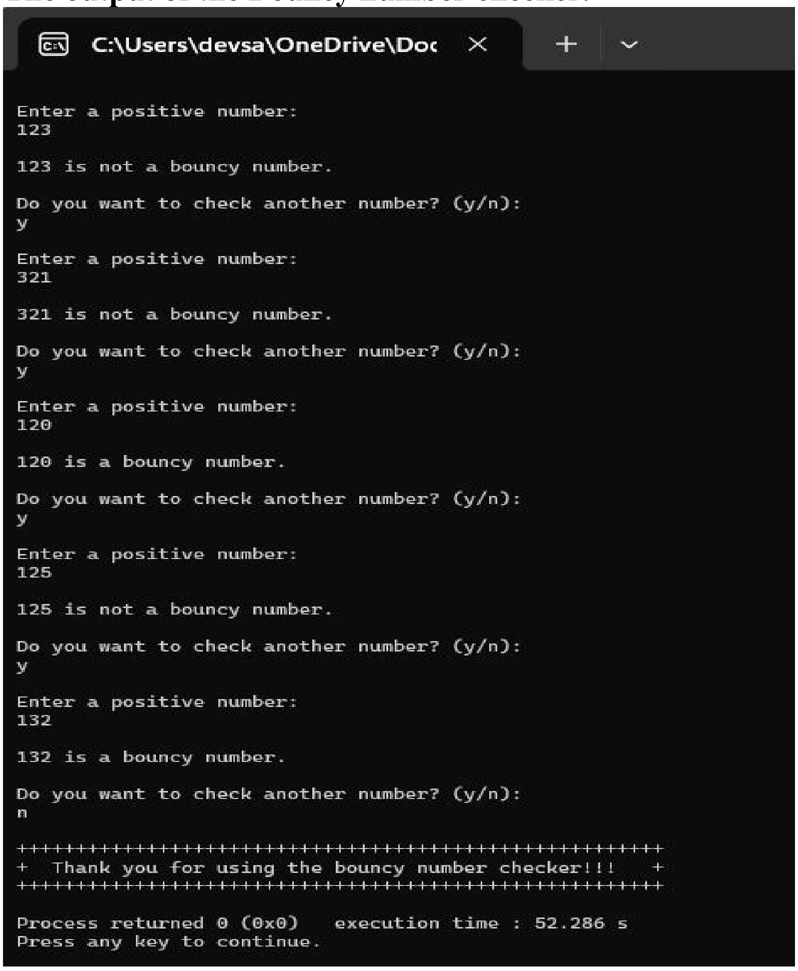
```
            else {
                cout << " " << endl;
                cout << n << " is not a bouncy number." << endl;
            }
        }

        cout << " " << endl;
        cout << "Do you want to check another number? (y/n): " << endl;
        cin >> choice;

    } while (choice == 'y');

    cout << " " << endl;
    cout <<
"++++++++++++++++++++++++++++++++++++++++++++++++++++++++++" <<
endl;
    cout << "+  Thank you for using the bouncy number checker!!!   +" << endl;
    cout <<
"++++++++++++++++++++++++++++++++++++++++++++++++++++++++++" <<
endl;
    return 0;
}
```

The output of the Bouncy number checker.



QUESTION NO.6

Write a program that manages a cinema ticket booking system. The program should display a 5x5 seating arrangement where:

- Available seats are marked with 'O'
- Booked seats are marked with 'X'

Program should:

- Display the current seating arrangement
- Ask user for row and column number (1-5) for booking
- Mark that seat as booked ('X')
- Show updated seating after each booking
- Display error if user selects already booked seat
- Display error if user enters invalid row/column numbers

SOLUTION:

```
#include <iostream>
using namespace std;

int main()
{
  char seat[5][5] = { {'O', 'O', 'O', 'O', 'O'},
                      {'O', 'O', 'O', 'O', 'O'},
                      {'O', 'O', 'O', 'O', 'O'},
                      {'O', 'O', 'O', 'O', 'O'},
                      {'O', 'O', 'O', 'O', 'O'} };

                      int row, col;
                      char choice;

                      do {
                         cout << "    Cinema Screen    " << endl;
                         cout << "--------------------" << endl;
                         for (int i = 0; i < 5; i++) {
                            for (int j = 0; j < 5; j++){
                               cout << seat[i][j] << " ";
                            }
                            cout << endl;
                         }
                         cout << "---------------------" << endl;

                         cout << "Enter a row to book a seat: ";
                         cin >> row;
                         cout << "Enter a column to book a seat: ";
                         cin >> col;

                         if (row < 1 || row > 5 || col < 1 || col > 5){
                            cout << " " << endl;
```

```
                        cout << "Invalid row or column. Please enter values between 1 and 5." << endl;
                    }
                    else if (seat[row - 1][col - 1] == 'X') {
                        cout << " " << endl;
                        cout << "Sorry! This seat is already booked." << endl;
                    }
                    else {
                        seat[row - 1][col - 1] = 'X';
                        cout << " " << endl;
                        cout << "Seat successfully booked!" << endl;
                    }
                    cout << " " << endl;
                    cout << "Updated seating arrangement: " << endl;

    cout << "Cinema Screen" << endl;
    cout << "--------------------" << endl;
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++) {
            cout << seat[i][j] << " ";
        }
        cout << endl;
    }
    cout << "---------------------------" << endl;

    cout << " " << endl;
    cout << "Do you want to book another seat? (y/n): ";
    cin >> choice;




                 } while (choice == 'y');
                cout << " " << endl;
                cout << "Thank you for booking!" << endl;
                return 0;
}
```

The output of the cinema ticket booking system is given below.

C:\Users\devsa\OneDrive\Doc

```
    Cinema Screen
--------------------
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
---------------------
Enter a row to book a seat: 1
Enter a column to book a seat: 1

Seat successfully booked!

Updated seating arrangement:
Cinema Screen
---------------------
X 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
----------------------------


Do you want to book another seat? (y/n): y
    Cinema Screen
--------------------
X 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
---------------------
Enter a row to book a seat: 2
Enter a column to book a seat: 2

Seat successfully booked!

Updated seating arrangement:
Cinema Screen
---------------------
X 0 0 0 0
0 X 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
----------------------------


Do you want to book another seat? (y/n): y
```

```
    Cinema Screen
--------------------
X O O O O
O X O O O
O O O O O
O O O O O
O O O O O
---------------------
Enter a row to book a seat: 1
Enter a column to book a seat: 1

Sorry! This seat is already booked.

Updated seating arrangement:
Cinema Screen
---------------------
X O O O O
O X O O O
O O O O O
O O O O O
O O O O O
----------------------------

Do you want to book another seat? (y/n): y
    Cinema Screen
--------------------
X O O O O
O X O O O
O O O O O
O O O O O
O O O O O
---------------------
Enter a row to book a seat: 6
Enter a column to book a seat: 6

Invalid row or column. Please enter values between 1 and 5.

Updated seating arrangement:
Cinema Screen
---------------------
X O O O O
O X O O O
O O O O O
O O O O O
O O O O O
----------------------------

Do you want to book another seat? (y/n): y
    Cinema Screen
--------------------
X O O O O
O X O O O
O O O O O
O O O O O
O O O O O
---------------------
Enter a row to book a seat: 5
Enter a column to book a seat: 4

Seat successfully booked!

Updated seating arrangement:
Cinema Screen
---------------------
X O O O O
O X O O O
O O O O O
O O O O O
```

My Github Link: https://github.com/sd6012/cpp_Assignment